

SUPPORTING INFORMATION

Isotopic exchange study on the kinetics of Fe carburization and the mechanism of the Fischer-Tropsch reaction

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Section I. Supplementary figures

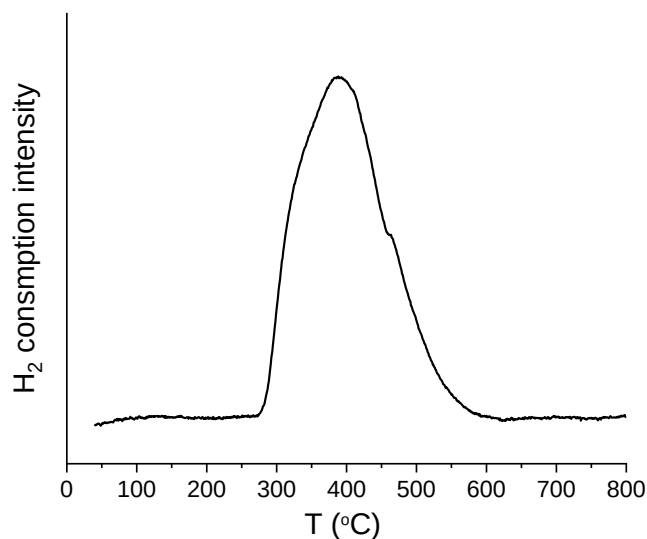


Figure S1. H₂-TPR profile of as-prepared Raney-Fe.

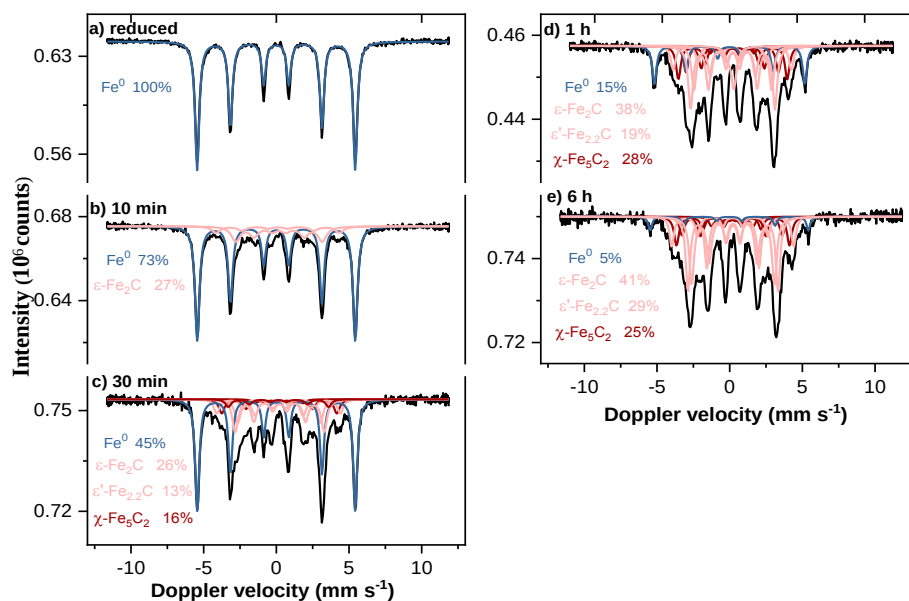


Figure S2. *In situ* Mössbauer spectroscopy of Raney-Fe catalyst. (a) Reduced at 20% H₂/Ar, 1 bar, 430 °C for 1 h; (b) Carburized at H₂/CO = 2, 1 bar, 250 °C for 10 min; (c) Carburized at H₂/CO = 2, 1 bar, 250 °C for 30 min; (d) Carburized at H₂/CO = 2, 1 bar, 250 °C for 1 h; (e) Carburized at H₂/CO = 2, 1 bar, 250 °C for 6 h.

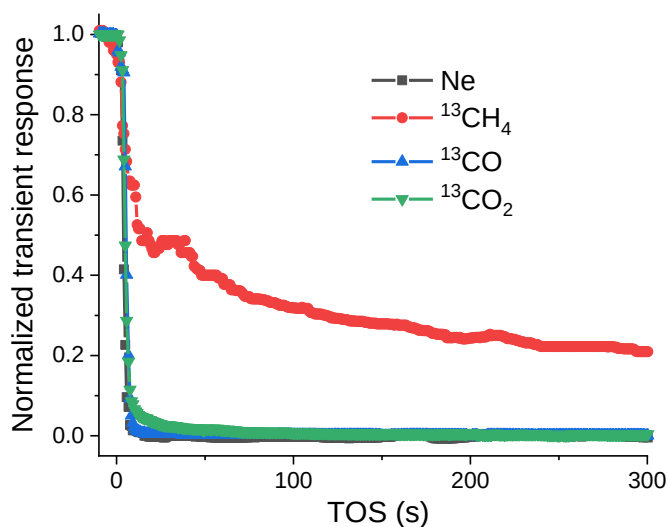


Figure S3. Transient response of Ne, ^{13}CO , $^{13}\text{CO}_2$, and $^{13}\text{CH}_4$ after a switch from $^{13}\text{CO}/\text{H}_2$ to $^{12}\text{CO}/\text{H}_2$ after 1 h carburization (8% CO + 16% H_2 , 250 °C, 1.5 bar).

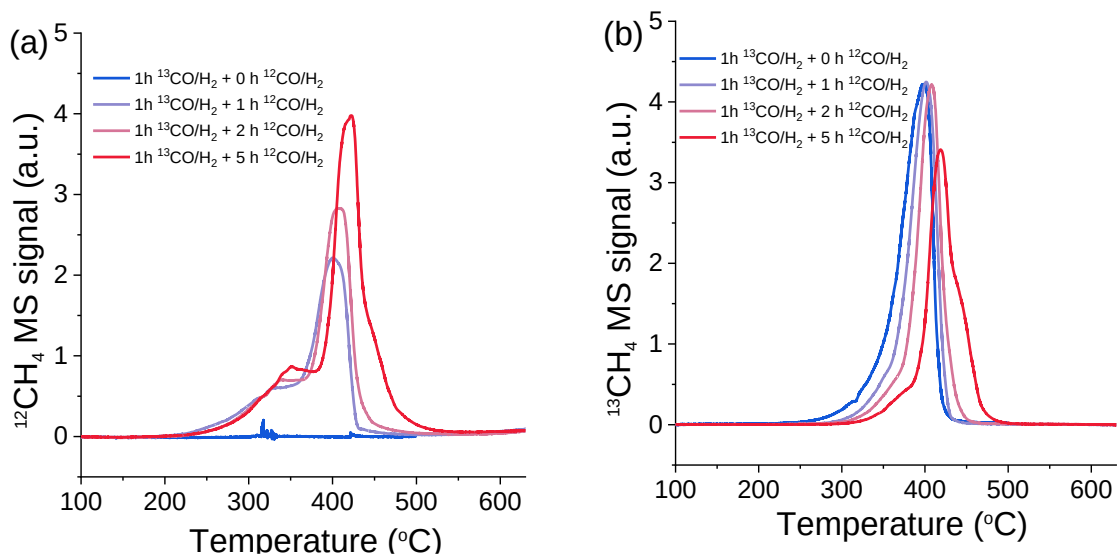
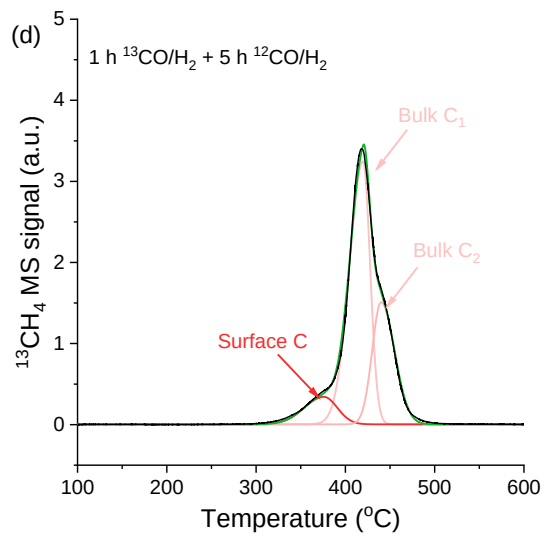
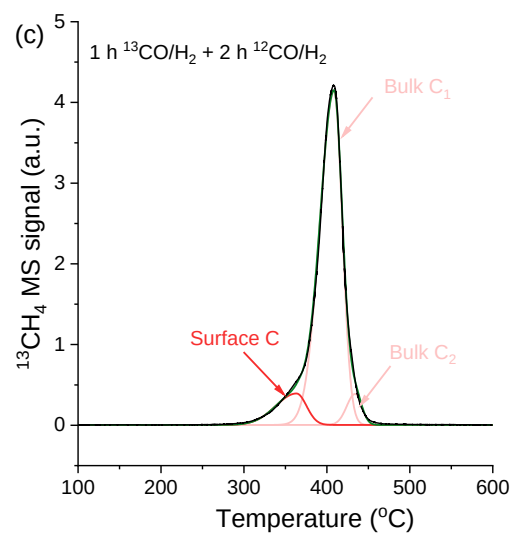
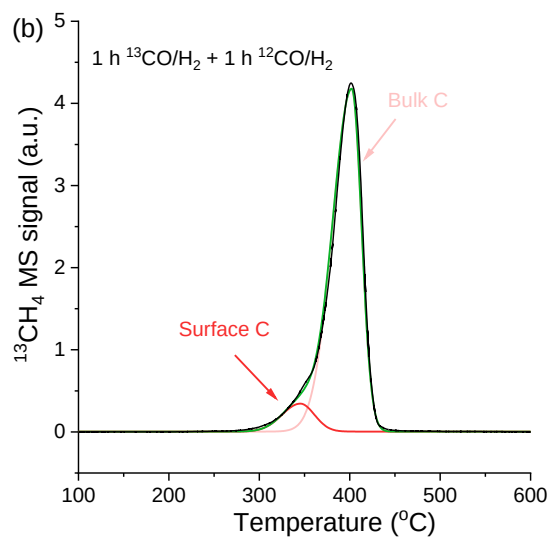
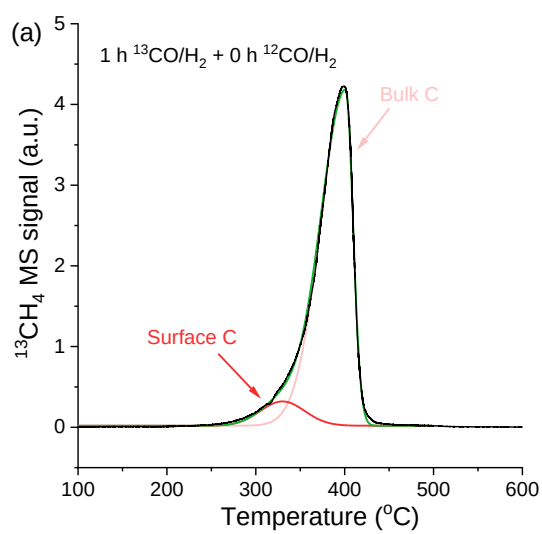


Figure S4. TPH profiles for (a) $^{12}\text{CH}_4$ and (b) $^{13}\text{CH}_4$ component in carburized samples with 1 h $^{13}\text{CO}/\text{H}_2$ followed by x h $^{12}\text{CO}/\text{H}_2$ (8% CO + 16% H_2 , 250 °C, 1.5 bar).



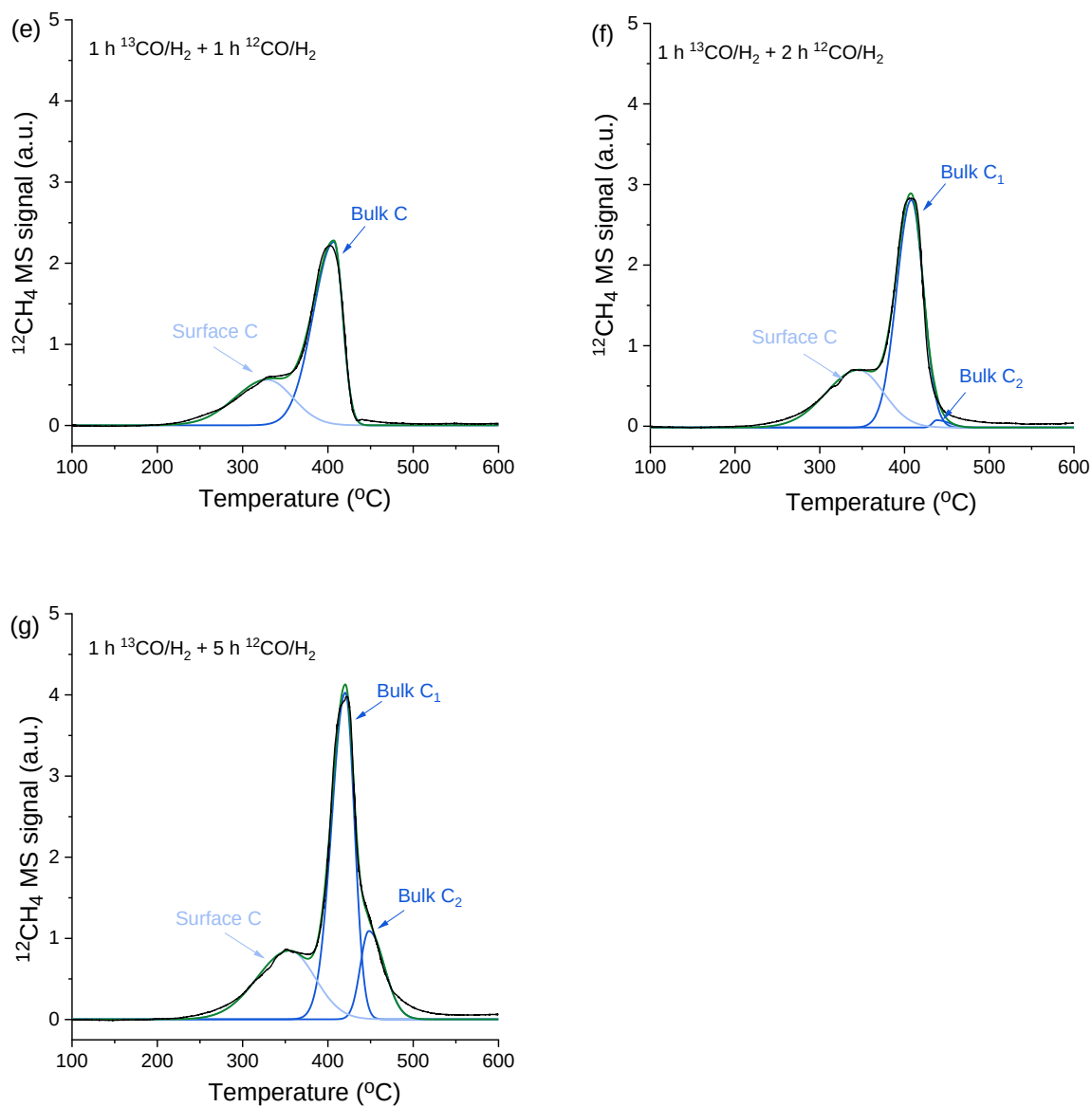


Figure S5. Deconvolution of (a-d) $^{13}\text{CH}_4$ TPH profiles after carburization in (a) 1 h $^{13}\text{CO}/\text{H}_2$, (b) 1 h $^{13}\text{CO}/\text{H}_2$ + 1 h $^{12}\text{CO}/\text{H}_2$, (c) 1 h $^{13}\text{CO}/\text{H}_2$ + 2 h $^{12}\text{CO}/\text{H}_2$, (d) 1 h $^{13}\text{CO}/\text{H}_2$ + 5 h $^{12}\text{CO}/\text{H}_2$ and (e-g) $^{12}\text{CH}_4$ TPH profiles after carburization in (e) 1 h $^{13}\text{CO}/\text{H}_2$ + 1 h $^{12}\text{CO}/\text{H}_2$, (f) 1 h $^{13}\text{CO}/\text{H}_2$ + 2 h $^{12}\text{CO}/\text{H}_2$, (g) 1 h $^{13}\text{CO}/\text{H}_2$ + 5 h $^{12}\text{CO}/\text{H}_2$ (8% CO + 16% H_2 , 250 $^{\circ}\text{C}$, 1.5 bar).

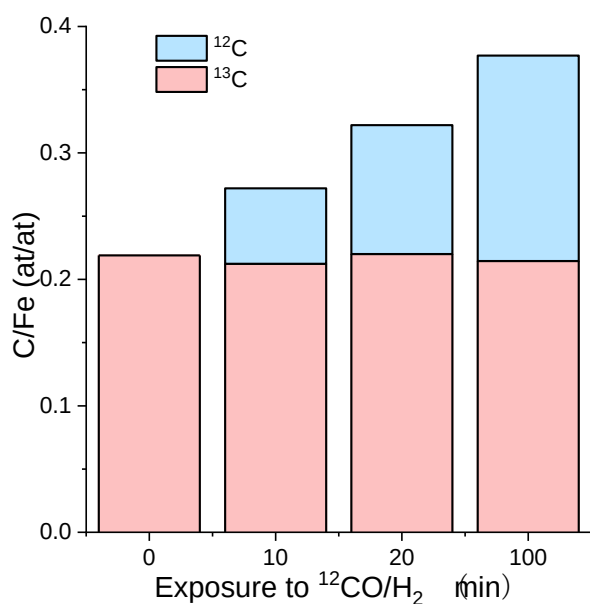


Figure S6. C/Fe ratio after carburization in $^{13}\text{CO}/\text{H}_2$ for 20 min followed by x min $^{12}\text{CO}/\text{H}_2$ (8% CO + 16% H_2 , 250 °C, 1.5 bar).

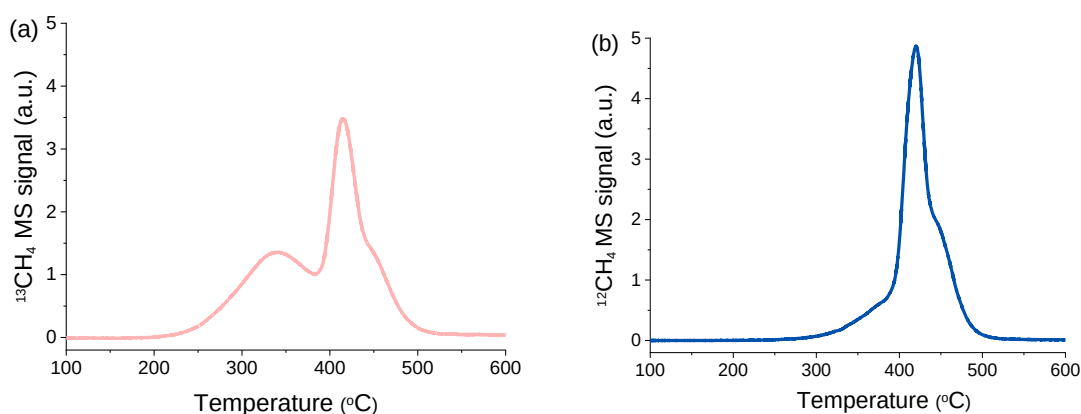


Figure S7. TPH profiles for (a) $^{13}\text{CH}_4$ and (b) $^{12}\text{CH}_4$ in samples carburized in $^{12}\text{CO}/\text{H}_2$ for 5 h followed by 1 h $^{13}\text{CO}/\text{H}_2$ (8% CO + 16% H_2 , 250 °C, 1.5 bar).

Section II. SSITKA data analysis

The transient $^{12}\text{CH}_4$ response was first corrected for catalyst deactivation. The deactivation was assumed to follow an exponential decay behavior due to the loss of active sites. It is assumed that the deactivation behavior before and after switch is the same. Therefore, determining the deactivation rate prior to the switch gives a measure for the correction factor of the transient response after the switch. k_{deact} before the switch can be obtained by Equation 1, in which R_0

represents the reaction rate of a fresh catalyst and R_t the reaction rate over time before the switch. It should be mentioned that k_{deact} is not a variable used in the fitting procedure.

$$R_t = R_0 \cdot e^{(-k_{\text{deact}} \cdot t)} \quad (\text{Eq. 1})$$

The reaction rate r after the switch can be described as a product of steady state rate $\bar{r}$ at the moment of the switch ($t = 0$), the transient response F , and exponential deactivation evolution (Equation 2).

$$r = \bar{r} \cdot F \cdot e^{(-k_{\text{deact}} \cdot t)} \quad (\text{Eq. 2})$$

$$F = \frac{r / \bar{r}}{e^{(-k_{\text{deact}} \cdot t)}} \quad (\text{Eq. 3})$$

Thus, the transient response after the switch can be corrected by k_{deact} (Equation 3). The corrected transient response was subsequently analyzed following previously described procedures for transient data treatment [1-3].

The formation rate of $^{12}\text{CH}_4$ is assumed to be pseudo-first order in the coverage of CH_x (θ_{CH_x}):

$$r_{\text{CH}_4} = k_{\text{CH}_x} \theta_{\text{CH}_x} \quad (\text{Eq. 4})$$

where k_{CH_x} is a pseudo-first order rate constant with unit s^{-1} .

The isotopic switch will lead to an exponential decay of the $^{12}\text{CH}_4$ signal. The transient response over time is given in Figure S7. It is evident that the signal cannot be described by a single exponential factor. We attribute this to the presence of multiple active sites with different intrinsic activities in CH_4 formation.

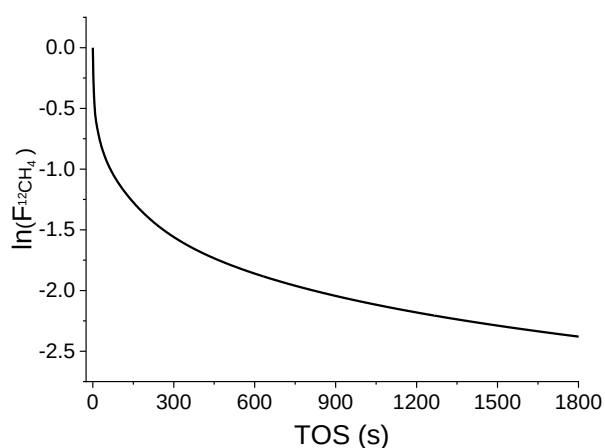


Figure S8. Semi-logarithmic plot of the normalized transient response of $^{12}\text{CH}_4$ over time from $^{12}\text{CO}/\text{H}_2$ to $^{13}\text{CO}/\text{H}_2$ after carburized in $^{12}\text{CO}/\text{H}_2$ for 5 h (8% CO + 16% H_2 , 250 °C, 1.5 bar).

The normalized transient response of the $^{12}\text{CH}_4$ after the isotopic switch, F , can be expressed as:

$$F = \sum x_i e^{(-k_i t)} \quad (\text{Eq. 5})$$

Each path rate r_i contributes x_i to the overall reaction rate r with a pseudo-first-order rate constant k_i :

$$x_i = \frac{r_i}{r} \quad (\text{Eq. 6})$$

The parameter sets x_i and k_i can then be determined by non-linear least square fitting. Figure 8 shows the transient response of $^{12}\text{CH}_4$ after a switch from $^{12}\text{CO}/\text{H}_2$ to $^{13}\text{CO}/\text{H}_2$ in combination with a line fitted by a varying number of exponential decay functions. The fitted rate constants and their fractional contribution to the overall rate are listed in Table S1. As can be seen, increasing the number of the reaction paths leads to a better fit as the value of coefficient of determination (R^2) approaches 1. The use of 4 paths yields a reasonably good fitting result. Further increasing the number of the reaction paths 4 hardly reduces the fitting error but increases the number of unknown parameters. Therefore, 4 paths have been used for fitting the SSITKA results. The reciprocal of the k_i indicates the residence time τ_i of the intermediate in the i th path. So the amount of the active site, N_i , is

$$N_i = x_i r \tau_i = x_i r k_i^{-1} \quad (\text{Eq. 7})$$

And the total amount of the active site, N , is

$$N = \sum N_i = r \sum x_i k_i^{-1} \quad (\text{Eq. 8})$$

For the i th path, the fraction number of the active sites y_i , can be calculated as

$$y_i = \frac{N_i}{N} = \frac{x_i k_i^{-1}}{\sum x_i k_i^{-1}} \quad (\text{Eq. 9})$$

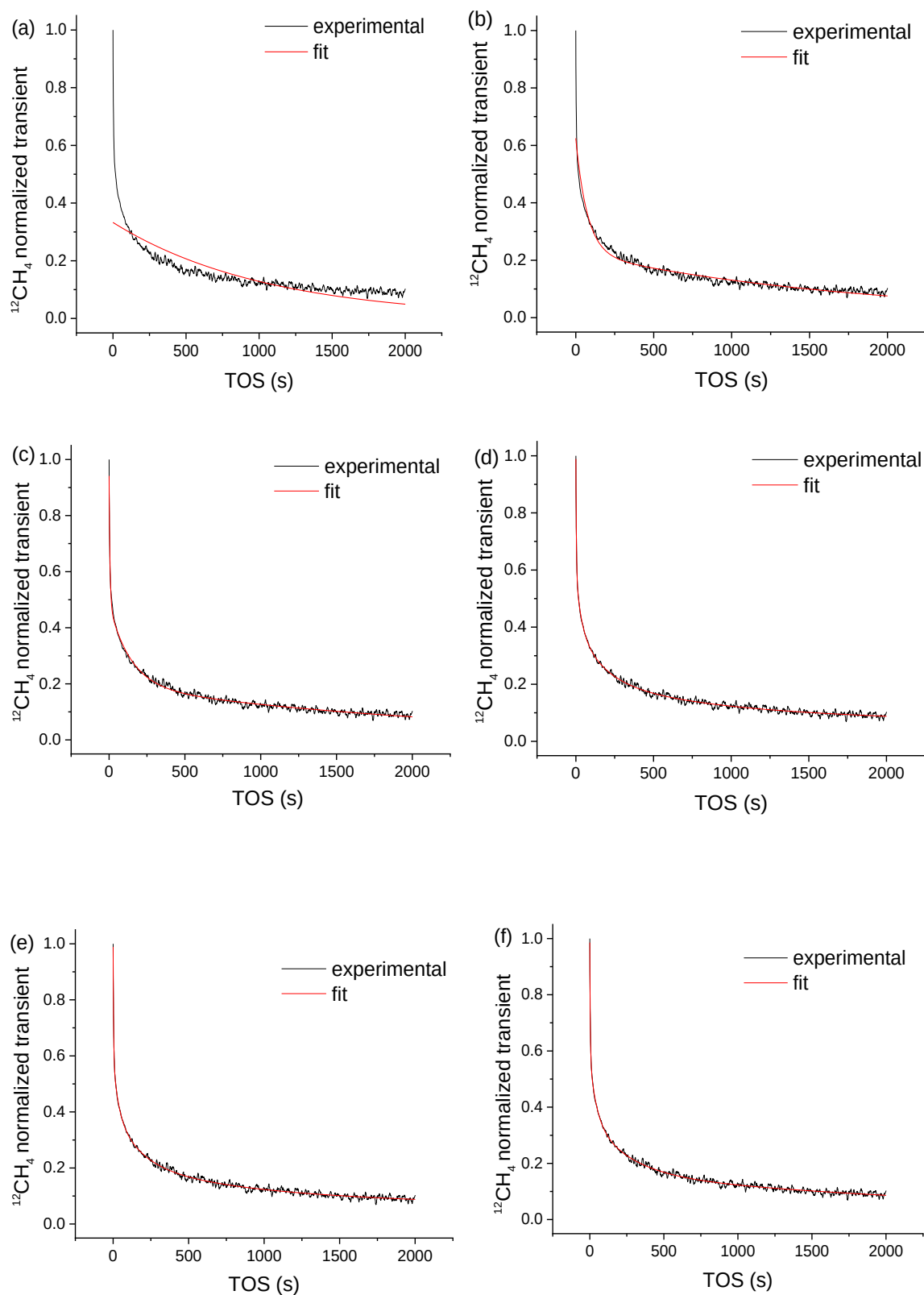


Figure S9. The normalized transient response of $^{12}\text{CH}_4$ after a switch from $^{12}\text{CO}/\text{H}_2$ to $^{13}\text{CO}/\text{H}_2$, fitted by (a) 1 path, (b) 2 paths, (c) 3 paths, (d) 4 paths, (e) 5 paths and (f) 6 paths.

Table S1. Fitted Parameters with an Increasing Number of Parallel Reaction Paths.

number of paths	rate constant k_i (s ⁻¹)	activity contribution x_i	sum of activity contributions	R ²
1	$9.52 \cdot 10^{-4}$	0.330	0.330	0.7285
2	$5.46 \cdot 10^{-4}$	0.225	0.624	0.9602
	$1.32 \cdot 10^{-2}$	0.399		
3	$4.29 \cdot 10^{-4}$	0.192	0.938	0.9892
	$7.23 \cdot 10^{-3}$	0.289		
	$1.65 \cdot 10^{-1}$	0.457		
4	$3.23 \cdot 10^{-4}$	0.165	0.985	0.9915
	$4.23 \cdot 10^{-3}$	0.199		
	$2.15 \cdot 10^{-2}$	0.208		
	$2.76 \cdot 10^{-1}$	0.413		
5	$1.01 \cdot 10^{-4}$	0.094	0.989	0.9916
	$1.20 \cdot 10^{-3}$	0.131		
	$6.12 \cdot 10^{-3}$	0.184		
	$2.96 \cdot 10^{-2}$	0.178		
	$2.94 \cdot 10^{-1}$	0.402		
6	$8.33 \cdot 10^{-5}$	0.089	0.988	0.9916
	$1.18 \cdot 10^{-3}$	0.066		
	$1.17 \cdot 10^{-3}$	0.068		
	$6.13 \cdot 10^{-3}$	0.185		
	$2.95 \cdot 10^{-2}$	0.178		
	$2.94 \cdot 10^{-1}$	0.402		

References

- [1] S. L. Shannon, J. Goodwin, Characterization of catalytic surfaces by isotopic-transient kinetics during steady-state reaction, Chem. Rev. 95 (1995) 677-695.
- [2] N. Lohitharn, J. Goodwin, Impact of Cr, Mn and Zr addition on Fe Fischer–Tropsch synthesis catalysis: Investigation at the active site level using SSITKA, J. Catal. 257 (2008) 142-151.
- [3] J. Yang, E. Z. Tveten, D. Chen, A. Holmen, Understanding the effect of cobalt particle size on Fischer–Tropsch synthesis: surface species and mechanistic studies by SSITKA and kinetic isotope effect, Langmuir 186 (2010) 16558-16567.